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METHOD OF STATEMENT FOR AIR CONDITIONING & MECHANICAL VENTILATION (ACMV) SYSTEM TESTING, COMMISSIONING, AND AIR BALANCING

DUCTED FAN COIL UNIT (FCU) SYSTEM AIR BALANCING PROCEDURE (REF.PIPE SYSTEM)

Prepared BY: _____
Subcontractor (Name & Signature) (Date)

Checked BY: _____
Project(M&E) (Name & Signature) (Date)
In- Charge

Approval BY: _____
Consultant (Name & Signature) (Date)

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1.0 Objective

This document provides a detailed, systematic procedure for the air balancing, testing, and commissioning of ducted Variable Refrigerant Flow (VRF) or Direct Expansion (DX) Fan Coil Unit (FCU) systems. The primary goal is to provide auditable proof that the FCU, its associated refrigerant circuit, and the entire air distribution network are installed and performing in full accordance with the project design specifications, approved shop drawings.

2.0 Scope of Work

This Method Statement covers all on-site activities required to achieve a fully functional and balanced system, including:

1. Verification of pre-requisite conditions, including the integrity of the refrigerant system.
2. Systematic air-side balancing of the ductwork network connected to the FCU.
3. Functional testing and performance verification of the automated refrigerant control system.
4. Verification of local and central controller functions.
5. All associated safety procedures and documentation.

3.0 References

1. Project-Specific Approved Shop Drawings, Technical Specifications, and Equipment Schedules.
2. VRF System Manufacturer's Installation.
3. **Workplace Safety and Health Act (WSH Act)** and subsidiary legislation.

4.0 Responsibilities

1. **Project Manager:** Overall responsibility for the successful execution of the project, ensuring all resources are available.
2. **Site Engineer:** To provide technical guidance, review and approve the method statement and test reports. To witness and verify the testing and commissioning process.
3. **Commissioning Coordinator:** To plan, coordinate, and execute all testing and commissioning activities. To ensure that all procedures are followed and documented.
4. **ACMV Technicians:** To carry out the installation, pre-commissioning checks, and testing and commissioning works under the supervision of the Commissioning Site Engineer.
5. **BMS Controls Contractor:** To be present during the testing of control systems and to make any necessary adjustments to the BMS.
6. **Safety Personal:** To ensure that all on-site activities comply with the WSH Act and project-specific safety requirements.

5.0 Preparation / Equipment Required

1. Submission and approval of this method statement.
2. Ensure the Calibrated air capture hood is assembled correctly with the appropriate skirt size (e.g., 600mm x 600mm, 1200mm x 600mm) that matches the nominal size of the diffuser or grille being tested.
3. Calibrated Multi-function Meter (e.g., TSI VelociCalc 9600 series or equivalent)
4. Static Pressure Probe (with side-sensing ports)
5. Flexible, color-coded tubing
6. Multi Clamp Meter
7. Vane Anemometer or HOT wire Anemometer
8. **Risk Assessment** for this activity has been approved, and a **Permit-to-Work (PTW)** has been issued if required.
9. Drill with appropriate bit size (if test ports are not pre-installed)
10. Duct plugs or high-quality aluminium foil tape
11. Safe access equipment (e.g., ladder, mobile scaffold, scissor lift if necessary)
12. Appropriate hand tools (e.g., Manifold Gauge, VCD adjustment tools, Allen keys).
13. Laminated, Warning Signs.
14. Appropriate Personal Protective Equipment (PPE): Safety helmets, safety glasses, gloves, safety footwear.
15. Full-body harnesses and lanyards for any work at height.
16. Approval Drawings and Air Balancing Test Sheets for data recording.

 Air Capture Hood Instruments	 Vane Anemometer	 Multi-function Anemometer
 Multi Clamp Meter	 PPE	 Ladder

(This section in the final report would include clear, dated photographs of the specific calibrated instruments used on-site, showing their serial numbers and calibration stickers.)

6.0 Pre-Requisites

Commissioning shall not commence until the following are formally verified.

1. **Refrigerant System Integrity:** The entire VRF/DX system, including all indoor and outdoor units, has been successfully **pressure tested with Oxygen-Free Nitrogen (OFN)**, **evacuated to a deep vacuum** and **charged with the precise quantity of refrigerant**.
2. **System Communication:** The system has been energized and a full **"auto-addressing" cycle** has been completed successfully, with the outdoor unit confirming communication with all connected indoor units and controllers.
3. **Physical Completion:** The FCU, filters, ductwork, and condensate drain piping are fully installed, clean, and free of debris.
4. **Safety:** A specific **Risk Assessment** for this activity has been approved, and a **Permit-to-Work (PTW)** has been issued if required.
5. The Building Management System (BMS) is capable of commanding the FCU to start or Start the FCU fan system from the CRC or Remote controller Manually
6. The FCU is electrically powered, and all Volume Control Dampers (VCDs) are set to 100% open.
7. Calibrated testing equipment and air balancing test sheets are ready.

7.0 Step-by-Step Balancing Procedure:

Step 1: Initial System Start-up and Verification

- 1 Start the FCU fan system from the starter panel (LCP) Manually or BMS.
- 2 Visually and audibly inspect the FCU for any abnormal noise, vibration, or operational issues.
- 3 Confirm the fan is rotating in the correct direction as indicated by the manufacturer.
- 4 Ensure chilled water is circulating through the cooling coil and control valves are operational.

Step 2: Adjust Total Airflow (Setting the Fan)

For FCUs with a Speed Regulator or Thermostat:

1. Access the Speed regulator control interface (local or BMS).
2. While monitoring the live airflow measurement from the pitot tube traverse, slowly increase or decrease the operating frequency (Hz) of the Speed regulator.
 - **Set Fan Speed** on the FCU controller, set the fan to the speed specified in the design total airflow schedule (e.g., "High").
4. Continue adjusting until the measured total airflow is within +/- 5% of the design total airflow.
5. Record the final operating frequency (Hz) and the corresponding motor running amperage (Amps) on the test sheet.

Step 3: Proportional Balancing of Main Duct Branches

1. Once the total airflow at the FCU is correct, move to the main branches of the ductwork.
2. Identify the "index branch" – this is the branch that is receiving the lowest percentage of its design airflow (i.e., the path of most resistance). The damper for this branch should remain fully open.
3. Slightly close the main dampers on the other, non-index branches to throttle their airflow. This will increase pressure in the system and divert more air towards the index branch.
4. Repeat measurements and adjustments until all main branches are proportionally balanced, meaning they each receive a similar percentage of their respective design airflow.

Step 4: Balancing Terminal Devices (Diffusers and Grilles)

1. Move to the rooms served by the "index branch."
2. Using the air capture hood, measure and record the airflow from every supply air diffuser connected to the FCU.
3. Identify the "index diffuser" – the one with the lowest reading relative to its design value. The VCD serving this diffuser should remain fully open.
4. Add all individual readings to get the Total Measured Airflow. Compare this with the FCU's total design airflow. If the total is incorrect, adjust the fan speed and re-measure until the total is within tolerance (+/- 10%)
5. Start with the diffuser with the highest airflow percentage. Slowly close its VCD until the design flow is met. This diverts pressure to other outlets. Repeat for all other diffusers, leaving the index diffuser's damper fully open.
6. Adjust the VCDs for the other diffusers in the same branch, throttling them down until all diffusers are providing their design airflow within the acceptable **tolerance (+/- 10%)**.
7. Slowly close the Volume Control Damper (VCD) for this grille while watching the live reading on your air capture hood.
8. Once one branch is complete, move to the next branch and repeat the process until all diffusers in the entire system are balanced.

Step 5: Verify Return Air System and Final Checks

1. Perform a Multifunction meter or Air Capture hood on the return duct air grille to measure the total return airflow. Verify this against the design.
2. Measure the fresh air intake airflow to ensure it meets the requirements of SS 553.
3. Conduct a final check of the total supply airflow at the FCU to ensure it has not significantly changed during the balancing process.
4. Mark the final positions of all VCD and damper tighten with a levers and record all final readings on the test sheets.

7.2 Phase B: Refrigerant Performance Verification

1. **Initiate Full Load Test:** From the VRF central controller, activate the system-wide "Test Run" mode, commanding all units to operate at maximum cooling capacity.
2. **Performance Data Collection:** Allow the unit to run continuously for at least 15-20 minutes to achieve stable conditions. Using a calibrated digital thermometer, measure:
 - **Return Air Temperature (T-Return):** At the main return air grille.
 - **Supply Air Temperature (T-Supply):** At a representative supply air diffuser.
3. **Performance Analysis (Delta-T):** Calculate the temperature difference: $\Delta T = (T\text{-Return} - T\text{-Supply})$.
 - **Acceptance Criterion:** A healthy ΔT for a VRF/DX system is typically between **8°C and 14°C**. This confirms the Electronic Expansion Valve (EEV) is functioning and adequate refrigerant is flowing through the coil.
4. **System Diagnostics:** During the test run, continuously monitor the local and central controllers.
 - **Acceptance Criterion:** The system must operate without generating any error codes.
5. **Record Performance Data:** Document the final measured temperatures, calculated ΔT , and a confirmation of "No Errors Found" on the commissioning test sheet

8.0 Local Control & Operation

The LCP provides the essential means to operate the equipment under different scenarios, typically managed by a **Hand-Off-Auto (H-O-A)** selector switch:

8.1 Local Control Functional Test ("Hand" Mode)

Turn the H-O-A selector switch on the LCP front panel to **"HAND"**.

1. Press the local "START" button.
 - a. **Verification:** The fan/motor should start and run. The "RUN" indicator lamp should illuminate.
2. Using a clamp-on ammeter, measure the running current on all three phases at the motor terminals or VSD output. The readings should be balanced and not exceed the motor's Full Load Ampere (FLA) rating. Record these initial readings.
3. Press the local "STOP" button.
 - a. **Verification:** The fan/motor should stop. The "RUN" lamp should extinguish.
4. While the motor is running in "HAND" mode, press the **Emergency Stop** button.
 - a. **Verification:** The motor must stop immediately. The LCP may show a "TRIP" or "FAULT" status.
5. **Testing and Commissioning:** Verifying motor rotation and initial performance.
6. Reset the E-Stop and any fault on the panel before proceeding. If available

8.2 Status Indication

The LCP provides immediate, at-a-glance visual feedback on the equipment's status via indicator lamps on the panel door:

1. **RUN / ON Light (Typically Green):** Indicates that the motor is energized and running.
2. **STOP / OFF Light (Typically Red):** Indicates that the motor is stopped.
3. **TRIP / FAULT Light (Typically Amber/Yellow):** Indicates that the equipment has stopped due to a protective device tripping (e.g., motor overload, phase failure), alerting technicians to a problem.

9.0 Commissioning With BMS

1. **Point-to-Point Verification:** Verify all status points (Run, Fault/Trip) and command points (Start/Stop, Speed Control) from the LCP are correctly wired and appear on the BMS graphic.
2. **Remote Command Test:** Place the fan's LCP selector switch in "AUTO" mode.
 - Send a "Start" command from the BMS and verify the fan starts.
 - Send a "Stop" command and verify the fan stops.
3. **Status Feedback Test:**
 - While the fan is running, verify the "Run Status" on the BMS is "ON" or "Running."
 - Manually trip the motor's overload relay at the LCP. Verify the fan stops and the "Fault" or "Trip" status appears on the BMS. Reset the fault.
4. **VSD Speed Control:** If applicable, send a speed control signal (e.g., 50%) from the BMS and verify the VSD's output frequency changes accordingly.

9.1 Functional Testing

Perform a final functional test by running the system under its normal automated control sequence for to ensure stable and reliable operation without nuisance tripping.

10.0 Documentation and Handover

1. Throughout the process, record all readings, observations, and test results on the Commissioning Test Sheet for the system.
2. Any failures, discrepancies, or non-conformities shall be recorded on a "Punch List" for rectification.
3. Upon successful completion of all tests, the Commissioning Engineer and Supervisor will sign the test sheet, confirming the System is safe and fully functional.
4. The completed and signed documentation will be included in the final Testing & Commissioning report for handover.
5. Compile the report for submission and verification with Maincon M&E / RTO or Consultant.

11.0 Health and Safety

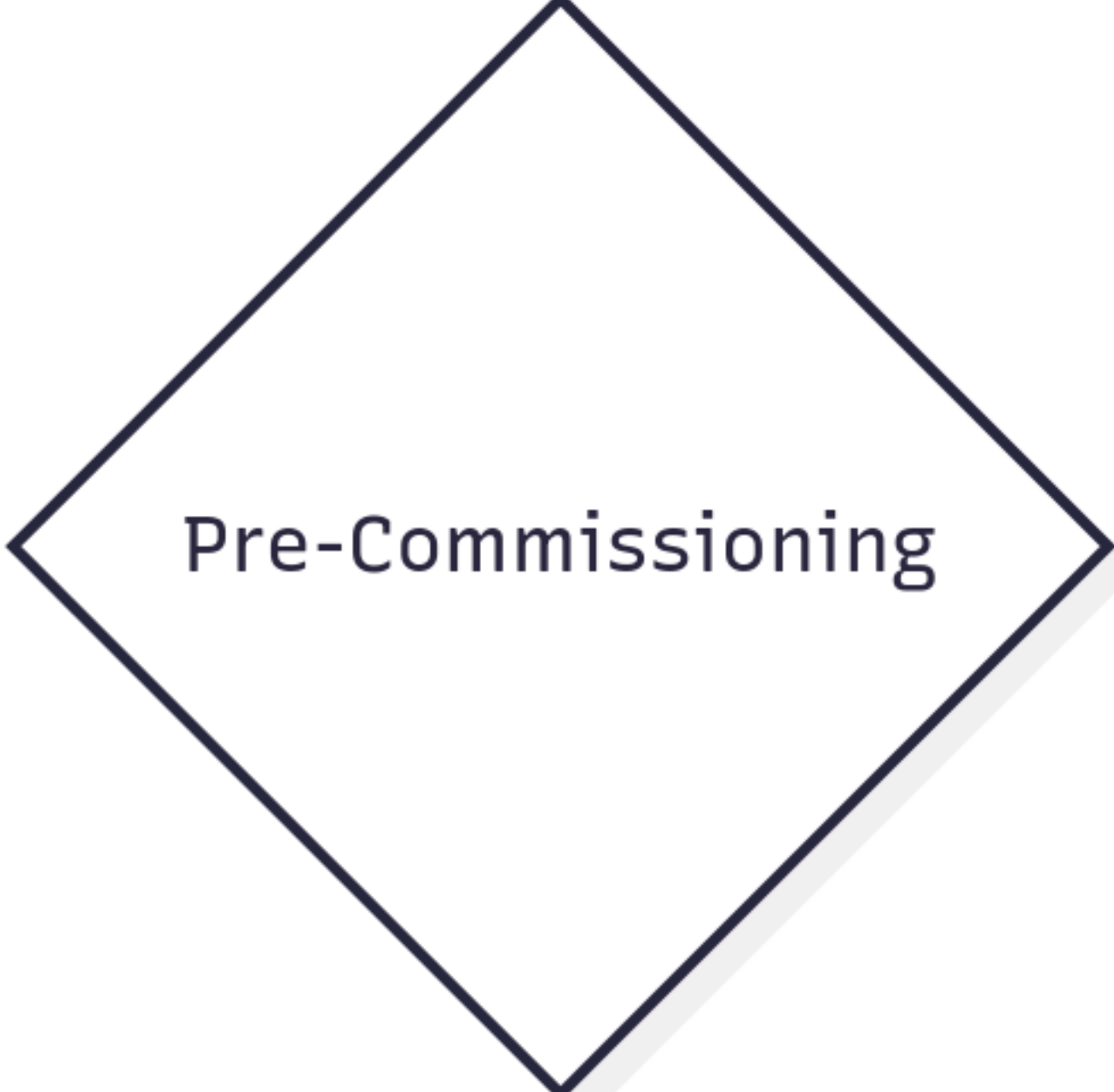
All work will be carried out in strict adherence to the project's Health, Safety, and Environment (HSE) plan and the WSH Act. Key safety measures include:

1. All personnel will wear the appropriate Personal Protective Equipment (PPE).
2. Lock-out/Tag-out (LOTO) procedures will be implemented before any work is carried out on energized equipment.
3. Safe access will be ensured for all work areas, especially at heights.
4. Good housekeeping will be maintained at all times.

13.0 APPENDIX

VRF FCU Testing Flowchart

Installation Complete



Check Piping, Wiring, Drains

Pressure & Leak Test
Passed?

No

Yes

Rectify Leaks

Evacuate & Charge
Refrigerant

Power Up & Initiate Test
Run

Check Unit Communication

Test Each Indoor Unit -
Cooling or Heating



No

Yes

Troubleshoot - Refer to
Mfg. Guide

Test Condensate Drains

Record Data

Handover

OPERATION DATA FOR INDOOR (SPLIT, VRV -AHU/FCU) UNIT			
PROJECT NAME			
LOCATION		INDOOR DESIGNATION	
AREA SERVED		INDOOR MODEL	
SYSTEM		MANUFACTURER	

S/No	DESCRIPTION	UNIT	TECHNICAL DATA
1	Total cooling capacity	kW	
2	Type of Refrigerant	-	
3	Motor Power	kW	
4	Power Supply	V/P/Hz	

S/No	COMPONENT	ITEM	UNIT	MEASURED DATA
1	Air on coil temperature (DB/WB)		°C	
2	Air off coil temperature (DB/WB)		°C	
3	System operating current	Starting current	Amp	
		Running current	Amp	
4	Supply air flow		CMH	
5	Return / Fresh Air Flow		CMH	
6	External static pressure		Pa	
7	Others			

Comments: (If there is no additional comments enter NONE)

Tested By		Witnessed By	Verified By
Signature			
Name			
Designation			
Company			
Date			



AIR BALANCING TEST REPORT

PROJECT NAME			
LOCATION		EQUIPMENT DESIGNATION	
AREA SERVED		EQUIPMENT MODEL	
SYSTEM		MANUFACTURER	

S/No	UNIT ID Tag	Room Designation	Grille Size	Design Air Flow (CMH)	Actual Air Flow (CMH)		Remarks
					Supply	Return	

Comments: (If there is no additional comments enter NONE)

Tested By		Witnessed By	Verified By
Signature			
Name			
Designation			
Company			
Date			

**Air Flow Testing & Commissioning Report**

Project Title :	Power Source (V/Ph/Hz):
Equipment Model:	FORMULA:(CMH = Average Velocity × (Duct or Grille Size) ×3600 CMH
Starting Ampere (A):	Running Ampere (A)
Location:	

No:	Equipment Designation	Area Served	Design CMH	Louver / Duct / Grille Size	P-1	P-2	P-3	P-4	P-5	Average Vel.M/s	Total CMH

Comments :-		
Tested By :	Witnessed By:	Verified By :
Signature :	Signature :	Signature :
Name :	Name :	Name :
Designation :	Designation :	Designation :
Company :	Company :	Company :
Date:	Date:	Date:

TEMPERATURE MEASUREMENT REPORT			
PROJECT NAME			
LOCATION		EQUIPMENT DESIGNATION	
AREA SERVED		EQUIPMENT MODEL	
SYSTEM		MANUFACTURER	

S/No	Room Designation	Temperature (°C)		Remarks
		Design	Measured	
			Dry /Wet	
1			/	
2			/	
3			/	
4			/	
5			/	
6			/	
7			/	
8			/	
9			/	
10			/	
11			/	
12			/	
13			/	
14			/	
15			/	

Comments: (If there is no additional comments enter NONE)

Tested By		Witnessed By	Verified By
Signature			
Name			
Designation			
Company			
Date			



GRACECAL PTE LTD (GST Reg No: 201611630E)

31 Woodlands Close #02-05 Woodlands Horizon Singapore 737855
Tel: (65) 6742 3850 | Fax: (65) 6742 3849 | Email: sales@gracecal.com
Website: www.gracecal.com

CALIBRATION CERTIFICATE

CLIENT : **WINNER ENGINEERING PTE LTD**
18 Woodlands Industrial Park E1
Singapore 757738

CERTIFICATE NO : GCC241442
ISSUE DATE : 06-SEP-2024

INSTRUMENT : **MULTI FUNCTION ANEMOMETER**
MAKE : TSI
MODEL NO. : TA465-P-NB
PART NO. : ---
SERIAL NO. : TA4652029012

WORKS ORDER NO : GWO241601
DATE RECEIVED : 02-SEP-2024

DATE CALIBRATED : 06-SEP-2024
DUE DATE : 06-SEP-2025

AMBIENT TEMPERATURE:
(23 ± 5) °C
RELATIVE HUMIDITY:
(55 ± 10) % r.h.

RANGE : -----
GRADUATION : -----
(TAG NO.) : -----
PAGE : 1 of 2
STATUS : AS FOUND

The described instrument has been calibrated at Laboratory under the ambient conditions stated above.

This certificate provides traceability of measurement to the International System of Units (SI) and/or to units of measurement realized at the National Metrology Centre (NMC), Singapore or other recognized national metrology institutes.

The Instrument was calibrated by comparison with laboratory calibration testing equipment.

1. Gas Deadweight Pressure Tester S/N: 558735

Calibration Method

Calibration Method is in accordance with ISO/IEC 17025. The calibration method was carried out according to Technical Calibration Procedure AS-TM-M03 as a guide.

Calibration Results

1. The results of calibration are given on the attached calibration data sheet(s).
2. The expanded uncertainty of measurement associated with the calibration is estimated at a confidence level of approximately 95% with a coverage factor of k=2.00.
3. The user should determine the suitability of the instrument for its intended use.

Calibrated By:

PREM

ID: 748656

Approved By:

KUMAR

ID: 748670





GRACECAL PTE LTD (GST Reg No: 201611630E)

31 Woodlands Close #02-05 Woodlands Horizon Singapore 737855
Tel: (65) 6742 3850 | Fax: (65) 6742 3849 | Email: sales@gracecal.com
Website: www.gracecal.com

CALIBRATION CERTIFICATE

CERTIFICATE NO : GCC241442
ISSUED DATE : 06-SEP-2024

WORKS ORDER NO : GWO241601
PAGE : 2 of 2

APPLIED ACTUAL VALUE (Pa)		MEAN INSTRUMENT READING (Pa)				
		Before Adj.	Ascending After Adj.	Error	Before Adj.	Descending After Adj.
Pressure Measurement						
0.00	0.1	---		-0.10	0.0	---
50.46	50.7	---		-0.24	50.7	---
477.53	477.8	---		-0.27	477.8	---
916.92	917.2	---		-0.28	917.2	---